

Macroeconomics III - Lecture 5a

Professor Guido Cozzi

April 28th, 2026

From "Endogenous" to "Truly Endogenous" growth

- In endogenous growth models we have seen so far (Chapters 8 and 9) long-term growth could be of only two types:
- **semi-endogenous**: policy could not affect long-run growth - discovered by Jones (*Journal of Political Economy*, 1995), Kortum (*Econometrica*, 1997), and Segerstrom (*American Economic Review*, 1998).
- **truly endogenous** with **scale effect**: larger population implied higher long-run growth - invented by Romer (*Journal of Political Economy*, 1995), and Aghion and Howitt (*Econometrica*, 1992)
- Now: **Truly endogenous** R&D-driven growth model **without scale effect**:
- policy can affect long-run growth
- larger population does not imply higher long-run growth

- As in Chapter 9, the final output of the R&D sector is: technological progress $= A_{t+1} - A_t$.
- The direct output of the R&D sector is: ideas.
- R&D is a labour-intensive production process! We assume it does not involve capital at all
- Inputs: the innovative effort of labour and the existing level of technology.

Basic Intuition - R&D Dilution

- The economy has many sectors
- R&D performed in one sector is often useful mainly for that sector
- Hence R&D per sector is what increases productivity in the economy
- New products, new firms, and new sectors arrive constantly
- They dilute the aggregate R&D across more and more sectors
- This **dilution effect** can eliminate the scale effect

Basic Intuition - R&D Dilution

- Scale-free truly endogenous growth theory claims that the number of sectors is proportional to population
- Hence R&D per sector follows R&D employment as a fraction of the labour force
- R&D employment as a fraction of the workforce is what causes productivity growth rate in the whole economy
- Small economies – Switzerland, Austria, Sweden - can in the long-run grow at the same rate as large economies – US, Japan, Germany - if they have the same fraction of the labour force employed in R&D
- This theory was developed relatively recently, by Smulders and Van de Klundert (*European Economic Review*, 1995), Dinopoulos and Thompson (*Journal of Economic Growth*, 1998), Peretto (*Journal of Economic Growth*, 1998), Young (*Journal of Political Economy*, 1998), Howitt (*Journal of Political Economy*, 1999)

A Scale-free Macromodel of Truly Endogenous Growth I

- A very simple variant of the model you have seen in Chapter 9
- The main difference is that the aggregate technological progress, $A_{t+1} - A_t$, increases in the R&D share, $s_{Rt} \equiv \frac{L_{At}}{L_t}$, of the labour force
- Instead of writing the aggregate production function of the R&D sector as in the semi-endogenous growth model of Chapter 9 as

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda, \quad (1)$$

which implied that the growth rate, g_t , would tend to

$$g^{se} = (1 + n)^{\frac{\lambda}{1-\phi}} - 1 \simeq \frac{\lambda n}{1 - \phi}$$

where superscript *se* stands for "semi-endogenous".

If we replace equation (1) with the following

$$A_{t+1} - A_t = \rho A_t s_{Rt}^\lambda, \quad (2)$$

A Scale-free Macromodel of Truly Endogenous Growth II

we obtain an entirely new model, which is immune to the scale effect problem.

- Notice: equation (2) is **not** the endogenous growth model you have seen in Chapter 9:

$$A_{t+1} - A_t = \rho A_t L_{A_t}^\lambda,$$

which just set $\phi = 1$ in equation (1).

- The new theory equation (2) is fundamentally different because it contains the ratio s_{Rt} instead of the level L_{A_t} !
- This guarantees that an increasing population and labour force will not increase the growth rate!
- In fact, let us assume a constant fraction $s_{Rt} = s_R$
- Equation (2) implies that an increase in L_t will generate an increase in the total R&D labour $L_{A_t} = s_R L_t$ without generating an increase in $A_{t+1} - A_t$

A Scale-free Macromodel of Truly Endogenous Growth III

- Therefore there is **no scale effect!**
- Moreover, since $\phi = 1$ we will have truly endogenous growth: policies permanently changing s_R will permanently change the growth rate of the technology

$$g_t = \frac{A_{t+1} - A_t}{A_t} = \rho s_R^\lambda \quad (3)$$

The Complete Model I

$$Y_t = K_t^\alpha (A_t L_{Yt})^{1-\alpha}$$

$$A_{t+1} - A_t = \rho A_t s_R^\lambda,$$

$$K_{t+1} = sY_t + (1 - \delta)K_t$$

$$L_{t+1} = (1 + n)L_t$$

$$L_{Yt} + L_{At} = L_t$$

$$L_{At} = s_R L_t$$

- Parameters: α , ρ , λ , δ , s , n , and s_R .
- Notice that s and s_R can be affected by policy
- State variables: K_t , L_t , and A_t .

- The technological growth rate is constant at

$$\frac{A_{t+1} - A_t}{A_t} = \rho s_R^\lambda \equiv g^e \quad (4)$$

where superscript e stands for "endogenous".

- Define $\tilde{k}_t \equiv K_t / A_t L_t$ and $\tilde{y}_t \equiv Y_t / A_t L_t$, which as in Chapter 9 gives $\tilde{y}_t = \tilde{k}_t^\alpha (1 - s_R)^{1-\alpha}$
- The transition of $\tilde{k}_t$ follows exactly that of Chapter 9:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g^e)} \left[s \tilde{k}_t^\alpha (1 - s_R)^{1-\alpha} + (1 - \delta) \tilde{k}_t \right] \quad (5)$$

- As in Chapter 9, $\tilde{k}_t$ will converge to its positive steady state value

$$\tilde{k}^* = \left(\frac{s}{n + g^e + \delta + ng^e} \right)^{\frac{1}{1-\alpha}} (1 - s_R)$$

and $\tilde{y}_t$ will converge to its positive steady state value

$$\tilde{y}^* = \left(\frac{s}{n + g^e + \delta + ng^e} \right)^{\frac{\alpha}{1-\alpha}} (1 - s_R)$$

- Consequently, percapita output $y_t = Y_t/L_t = \tilde{y}_t A_t$ will eventually grow at the same rate as technology A_t , that is:

-

$$\frac{y_{t+1} - y_t}{y_t} \xrightarrow{t \rightarrow \infty} \frac{A_{t+1} - A_t}{A_t} = \rho s_R^\lambda \equiv g^e \quad (6)$$

Implications III

- Notice that, unlike semi-endogenous growth theory, in this new truly endogenous growth theory without scale effect percapita GDP can be increasing also in the absence of population growth - that is when $n = 0$, or even with $n < 0$ (shrinking population)

Therefore, according to this theory, the long-run (steady state, or balanced growth path) growth rate of GDP can persistently be influenced by anything that can affect the fraction of the labour force employed in research and development activities:

R&D subsidies by the government stimulating private R&D investment and employment

Patent policies protecting innovations and increase the profitability of R&D

Public investment in R&D laboratories

Public universities

Scholarships in support of post-graduate students (Masters, PhD students)

Implications IV

- The US government has always spent a lot in basic and applied R&D activities, with important results on innovation¹
- The truly endogenous growth theory without scale effect can be recast in terms of R&D investment as a fraction of GDP, instead of the R&D employment as a fraction of GDP, obtaining the same long-run results
- Evidence confirms that long-run TFP growth is positively associated with different measures of R&D intensity also at the firm level
- Firms, within a sector, spending a larger share of their value added on R&D will be more innovative and successful
- It is a very robust and influential theory... as is semi-endogenous growth theory.

¹See for example, the recent book: Mariana Mazzucato, *The Entrepreneurial State*, 2013 (<https://marianamazzucato.com/entrepreneurial-state/>)

Summing Up I

- Two important endogenous growth theories are currently the most influential
- **Semi-endogenous without scale effect**
- **Truly endogenous without scale effect**
- Both eliminate the prediction of the simplistic truly endogenous growth model with scale effect initially introduced by Paul Romer (1990), that is the one with $\varphi = 1$ and $n = 0$ we have seen in Chapter 9
- They differ in their long-run policy implications
- Semi-endogenous predicts that policy can affect growth only **temporarily** (in the transition)
- Truly endogenous predicts that policy can affect growth **permanently**
- Both theories have strong empirical support

Summing Up II

- Unlike the semi-endogenous, the truly endogenous without scale effect does not predict that growth be increasing with the population growth rate n
- It is not yet clear which of the two is "right" and which is "wrong"
- Empirical estimates of φ tend to be lower but close to 1, so that not it is not yet possible to exclude that either of the two theories is right.